1 Numerical measures, graphs and diagrams

Subject content		Additional information
1.1	Interpret statistical diagrams including bar charts, stem and leaf diagrams, box and whisker plots, cumulative frequency diagrams, histograms (with either equal or unequal class intervals), time series and scatter diagrams.	Students will not be required to draw or construct statistical diagrams. Students may be required to comment on or interpret the main features of published visualisations (for example those listed opposite) or critically assess published visualisations. This may include examples which draw upon multivariate data.*
1.2	Know the features needed to ensure an appropriate representation of data using the above diagrams, and how misrepresentation may occur.	
1.3	Justify appropriate graphical representation and comment on those published.	
1.4	Compare different data sets, using appropriate diagrams or calculated measures of central tendency and spread: mean, median, mode, range, interquartile range, percentiles, variance and standard deviation.	Students may be required to comment on or interpret the main features of published visualisations (for example those listed in section 1.1). This may include examples which draw upon multivariate data.*
1.5	Calculate measures using calculators and manual calculation as appropriate.	Where raw data is given candidates will be expected to obtain the values of the mean, standard deviation and variance directly from a calculator.
1.6	Identify outliers by inspection and using appropriate calculations.	
1.7	Determine the nature of outliers in reference to the population and original data collection process.	
1.8	Appreciate that data can be misrepresented when used out of context or through misleading visualisation.	This may include examples which draw upon multivariate data.*

*Please see *Appendix 4: Use of calculators and other technology*.